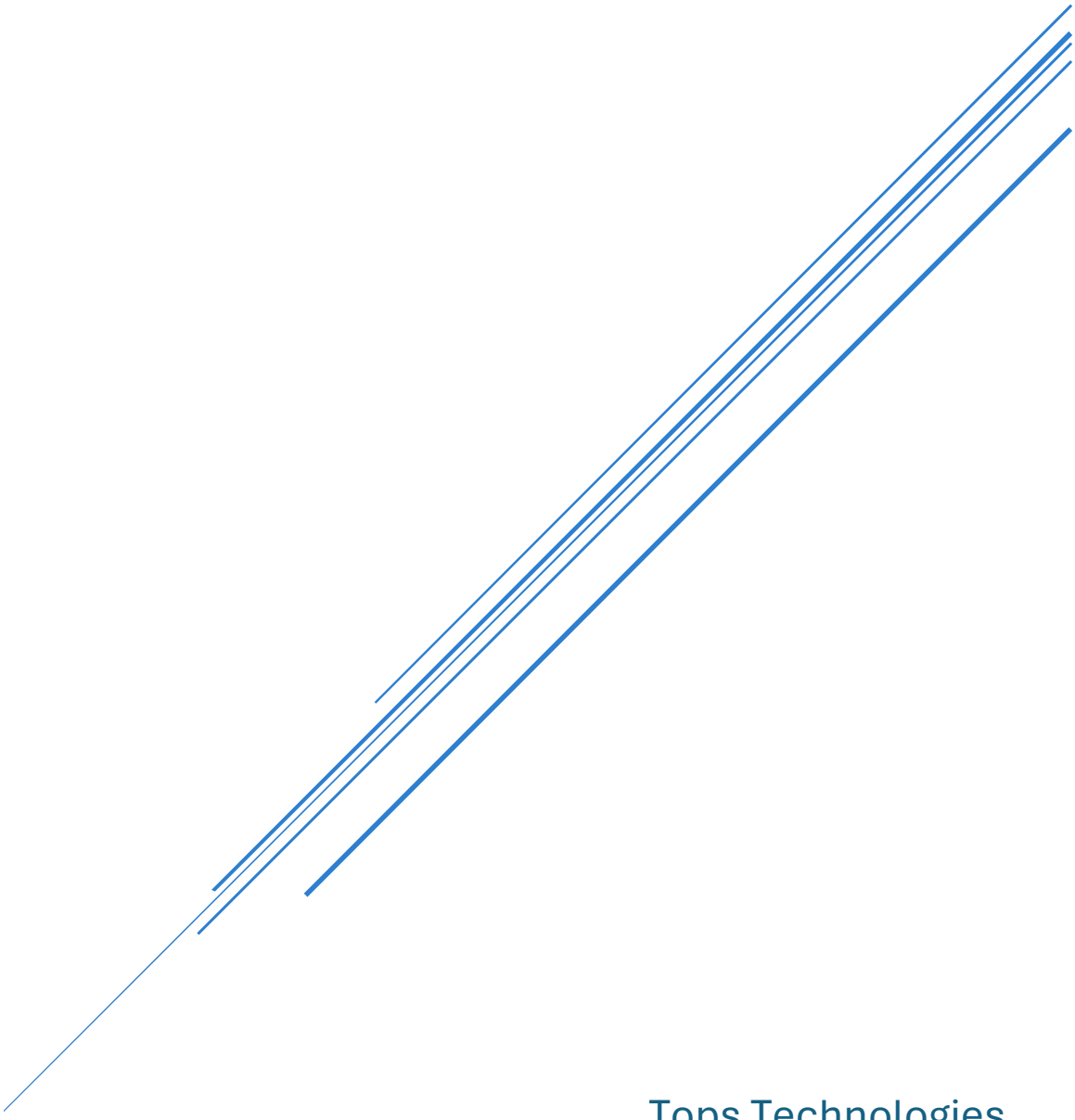
IP ADDRESSING & SUBNETTING

Assignment - 7



Tops Technologies
Cyber security

1. **List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.**

➢

➢ IP Address Classes (A, B, C, D, E)

IP Class	IP Address Range	Purpose	Real-World Example
Class A	1.0.0.0 – 126.255.255.255	Large networks	A large company's internal network using a private Class A address such as `10.0.0.1`.
Class B	128.0.0.0 – 191.255.255.255	Medium-sized networks	A university network using a private Class B address such as `172.16.0.1`.
Class C	192.0.0.0 – 223.255.255.255	Small networks	A home router using a private Class C address such as `192.168.1.1`.
Class D	224.0.0.0 – 239.255.255.255	Multicasting	A video-streaming or conferencing service sending data to a multicast group.
Class E	240.0.0.0 – 255.255.255.255	Experimental and reserved use	Researchers testing experimental networking protocols in a controlled environment.

2. **Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses. Hint: Show your working steps for each calculation.**

➢

Given:

- **IP Address: 192.168.10.45**
- **Subnet Mask: 255.255.255.0**

Step 1: Find the Network Address

Use the formula:

Network Address = IP Address AND Subnet Mask

IP Address	192	168	10	45
Subnet Mask	255	255	255	0
Result	192	168	10	0

The first three octets remain the same because the subnet mask is 255.255.255, and the last octet becomes 0.

Network Address: 192.168.10.0

Step 2: Find the Broadcast Address

The subnet mask 255.255.255.0 is /24. This leaves 8 host bits.

- **Host bits = \(32-24=8\)**
- **Total addresses = \(2^8=256\)**
- **Broadcast address = Network address + 255**

Therefore, the last octet is 255.

Broadcast Address: 192.168.10.255

Step 3: Find the Valid Host Address Range

- **First valid host = Network address + 1**
- **Last valid host = Broadcast address − 1**

So:

- **First valid host: 192.168.10.1**
- **Last valid host: 192.168.10.254**

Valid Host Range: 192.168.10.1 – 192.168.10.254

Final Answer

Item	Answer
IP Address	192.168.10.45
Subnet Mask	255.255.255.0 (/24)
Network Address	192.168.10.0
Broadcast Address	192.168.10.255
Valid Host Range	192.168.10.1 – 192.168.10.254
Total Addresses	256
Usable Host Addresses	254

3. **Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.**

➢

Subnetting Plan for a Food Delivery Startup Like Zomato

We are given the network 10.0.0.0/24 and need to create 4 separate subnets for Delivery Partners, Restaurants, Customers, and Admins.

Step 1: Calculate the new subnet mask

- **Original network: 10.0.0.0/24**
- **Required subnets: 4**
- **Borrowed bits: \(2^2 = 4\) subnets**
- **New prefix: /24 + 2 = /26**
- **New subnet mask: 255.255.255.192**
- **Addresses per subnet: \(2^{32-26}=64\)**
- **Usable host addresses per subnet: \(64-2=62\)**

Step 2: Subnetting table

Group	Network Address	Subnet Mask	Usable Host Range	Broadcast Address
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1 – 10.0.0.62	10.0.0.63
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65 – 10.0.0.126	10.0.0.127
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129 – 10.0.0.190	10.0.0.191
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193 – 10.0.0.254	10.0.0.255

Step 3: Verify the block size

Block size = 256-192=64

Therefore, the network addresses increase by 64:

- **First subnet: 10.0.0.0**
- **Second subnet: 10.0.0.64**
- **Third subnet: 10.0.0.128**
- **Fourth subnet: 10.0.0.192**

Each subnet provides 62 usable IP addresses, for a total of 248 usable host addresses across all four subnets.

4. **You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet.**
   Constraint: Show your calculation steps and final answers.

➢

Step 1: We need at least 20 subnets. \(2^4=16\) is too small, and \(2^5=32\) is enough, so we borrow 5 bits.

Step 2: New prefix = /16 + 5 = /21.

Step 3: Subnet mask = 255.255.248.0.

Step 4: Host bits = \(32-21=11\).

Hosts per subnet = \(2^{11}-2=2048-2=\mathbf{2046}\).

Step 5: Block size = \(256-248=8\). Therefore, the subnets increase by 8 in the third octet.

Subnet No.	Network Address	Valid Host Range	Broadcast Address
1	172.16.0.0/21	172.16.0.1 – 172.16.7.254	172.16.7.255
2	172.16.8.0/21	172.16.8.1 – 172.16.15.254	172.16.15.255
3	172.16.16.0/21	172.16.16.1 – 172.16.23.254	172.16.23.255
…	…	…	…
32	172.16.248.0/21	172.16.248.1 – 172.16.255.254	172.16.255.255

Final Answers

- **New subnet mask: 255.255.248.0 (/21)**
- **Total subnets: 32**
- **Usable hosts per subnet: 2046**

5. **A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.**

➢

The IP address 192.168.1.300 is invalid because each octet of an IPv4 address must be between 0 and 255. Here, 300 exceeds the maximum allowed value.

A valid IP address in the same subnet could be 192.168.1.10, assuming the subnet mask is 255.255.255.0 and the address is not already in use.